

Motivation

Empirical evidence: a positive relationship between elicited **risk preferences** and **time preferences**—greater risk aversion is often associated with stronger discounting (e.g., Dean & Ortoleva, 2019; Chapman et al., 2023).

→ A possible common factor: **cognitive ability** (Dohmen et al., 2010).

→ The observed risk–time correlation may partly reflect cognitive factors rather than related preferences.

- **Risk:** complexity can reproduce familiar lottery anomalies without actual risk (Oprea, 2024).
- **Time:** complexity can generate hyperbolic-like behavior without actual delay (Enke, Graeber, and Oprea, 2025).

A potential mechanism: complexity

Proper valuation requires costly information processing. Cognitively demanding tasks may instead induce reliance on imperfect decision procedures.

- **Attenuation:** insufficient responsiveness of valuations to task fundamentals.
- **Measurement implication:** the observed relationship between Risk and Time may depend on domain-specific responses to complexity.

This project moves beyond final choices:

Calculator-input data reveal the procedures used to construct valuations.

→ Key questions:

1. Stability of decision procedures across Risk and Time, or domain-specific?
2. Classification of decision makers into types based on their procedures?
3. Does computational complexity contribute to the observed correlation between elicited economic preferences?

Classifying Decision Procedures

Observed in Each Valuation Round

- the participant's final valuation;
- the complete sequence of calculator entries.

Calculator entries provide a direct, though incomplete, measure of how participants construct their valuations. (Zhang, 2025)

The calculator logs are classified according to the procedure visible in the participant's entries.

Procedure	Classification
Benchmark-based	Expected-value calculation in Risk, or exponential/repeated discounting in Time.
Number-only	Numbers entered without an identifiable structured calculation.
Mixed/heuristic	Alternative shortcuts, partial calculations, or procedures that change across trials.

From calculator entries to procedure classifications

Classification uses expressions recorded in the participant-visible calculation history.

A. Participant-visible calculator

Line	Calculation	Result
1	7*18/100	1.26

B. Illustrative coding of recorded entries

7*18/100
Result: 1.26 → Expected-value calculation

7*(0.95^8)
Result: 4.64 → Exponential/repeated discounting

5
Result: 5.00 → Number only

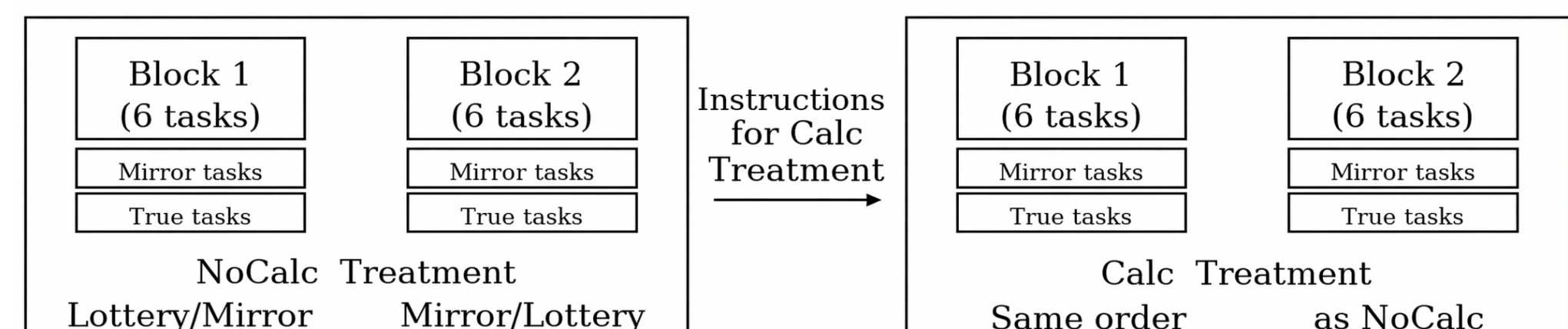
7-0.5*2
Result: 6.00 → Linear/subtractive adjustment

Illustrative examples; mental calculations not entered in the interface are not observed.

Experimental Design

- **Within-subject design** covering two domains: **Risk** and **Time**.
- Each domain contains two task structures: a **True task** and a computationally matched **Mirror task**.
- Participants complete the tasks first without and then with access to an embedded calculator.

	True task	Mirror task
Risk	Binary lottery	Deterministic average of the lottery outcomes
Time	Delayed monetary payment	Repeated shrink operation without an actual delay



- Participants first complete the **NoCalc treatment**.
- They then receive calculator instructions and repeat the same tasks in the **Calc treatment**.
- The task order is held fixed across NoCalc and Calc, allowing within-participant comparisons.

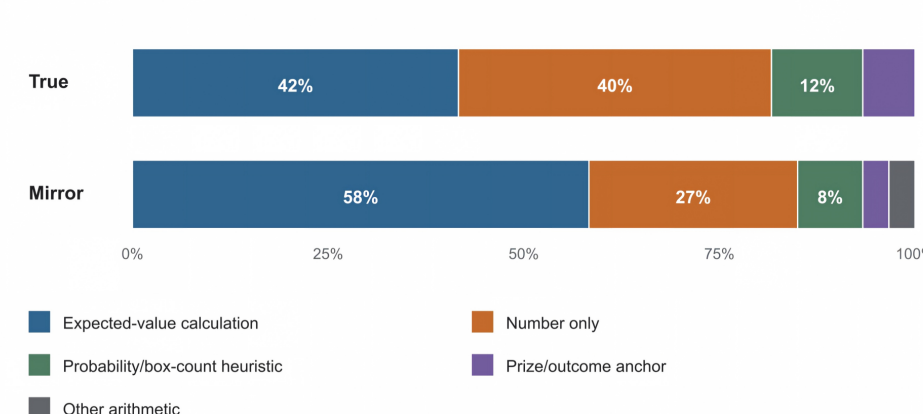
The findings shown below are based on the calculator pilot with $N = 30$ completed Prolific participants and 240 valuation rounds.

Pilot Results

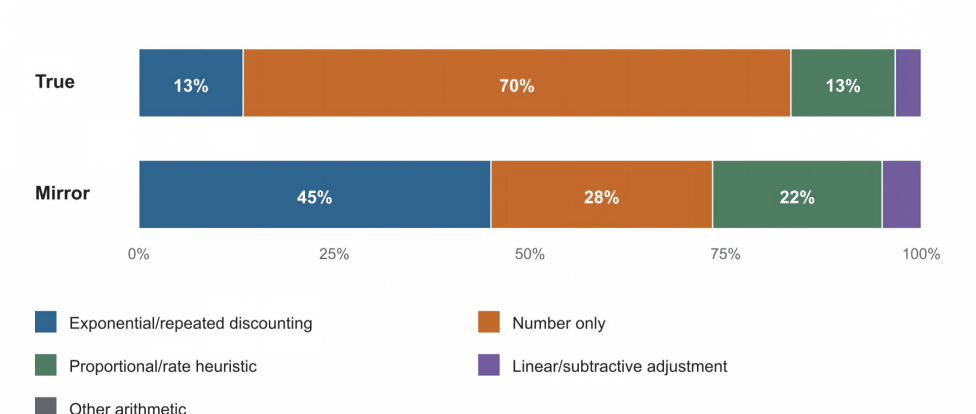
Calculator procedures vary by task type

Share of task trials; $N = 30$ participants (60 trials per bar)

A. Risk domain



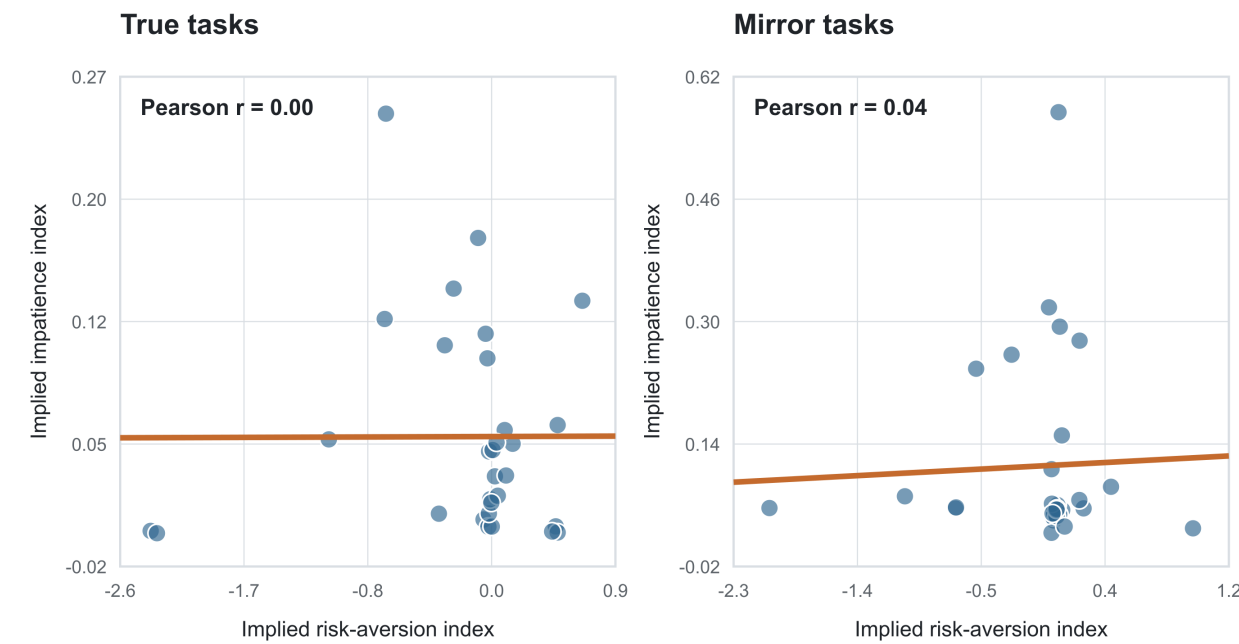
B. Time domain



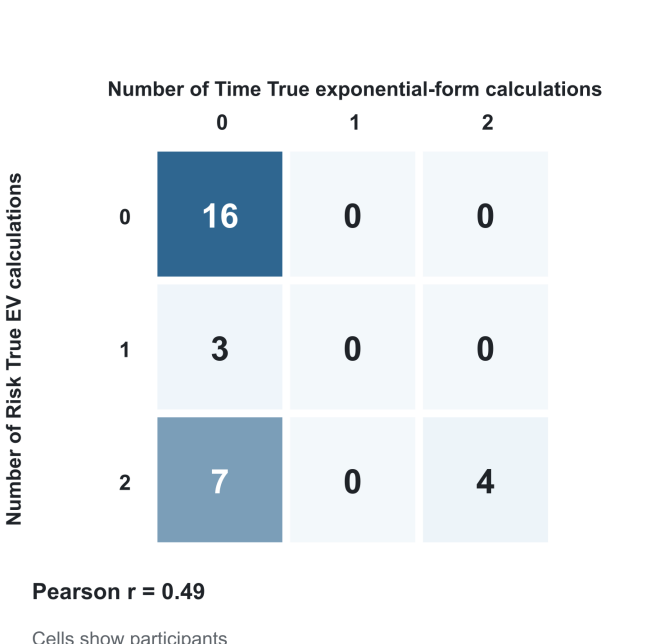
Cross-domain correspondence in valuations and observed procedures

Calculator pilot ($N = 30$)

A. Valuation-implied preference indices



B. Benchmark calculations in True tasks



Preliminary Takeaway

Task types changes observed procedures, especially in Time. Benchmark-calculation use shows modest cross-domain correspondence though elicited preferences do not.

Next steps:

- **Simplify design:** Assess if a True-task design better identifies calculator effects on valuations, sensitivity, heterogeneity, and the Risk–Time correlation.
- **Representation difficulty:** Test whether the Time domain involves greater representational difficulty than Risk.